Supplementary Material for

Lossless In-Domain, Lossy under Shift: Post-Training Quantization Can Make Tiny Machine Learning Wildfire Detectors Miss Fires

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Notation. TinyMWAMNet-S: MobileNetV3-Small at 128 pixels; TinyMWAMNet-S6: the same network with ReLU6; TinyMWAMNet-XS: MobileNetV1 of width 0.25 at 96 pixels. MNv1-0.25, MNv2-0.35 and MNv3-S denote MobileNetV1 (width 0.25), MobileNetV2 (width 0.35) and MobileNetV3-Small, followed by the input size in pixels. CE-std, CE-conf, KD-std and CAD are the four training regimes of Section 3.5 of the article. SD: standard deviation.

Table S1: Quantization-aware training without and with the range warm-up. INT8 cost: INT8 minus FP32 balanced accuracy (points, mean ± SD over seeds; five seeds for TinyMWAMNet-S, three for the size family). "None": the pre-specified schedule without warm-up (Notebook 03), scored with LiteRT's XNNPACK kernels; "warm-up": 3,000 range-estimation passes before QAT (Notebook 04), scored with LiteRT's built-in kernels.

		In-domain		**Shift-1**		**Shift-2**	
Model	**Regime**	**None**	**Warm-up**	**None**	**Warm-up**	**None**	**Warm-up**
TinyMWAMNet-XS (MNv1-0.25, 96)	CE-std	0.1 ± 0.2	0.3 ± 0.3	−0.1 ± 0.3	0.0 ± 0.4	−0.5 ± 0.6	−0.3 ± 0.8
TinyMWAMNet-XS (MNv1-0.25, 96)	CAD	0.0 ± 0.3	0.0 ± 0.0	−0.5 ± 0.3	−0.3 ± 0.1	0.3 ± 1.1	−0.1 ± 0.3
MNv3-S, 96	CE-std	−0.4 ± 0.4	−0.3 ± 0.3	−4.3 ± 0.6	−1.3 ± 0.6	−3.7 ± 0.7	−0.2 ± 2.2
MNv3-S, 96	CAD	−1.0 ± 1.0	0.1 ± 0.4	−3.1 ± 1.1	−0.6 ± 0.5	−9.6 ± 2.1	−0.2 ± 2.4
TinyMWAMNet-S (MNv3-S, 128)	CE-std	−0.1 ± 0.6	−0.2 ± 0.5	−4.1 ± 1.3	−1.5 ± 1.0	−2.9 ± 1.0	−0.7 ± 1.0
TinyMWAMNet-S (MNv3-S, 128)	CE-conf	−0.5 ± 0.3	−0.4 ± 0.1	−3.4 ± 1.3	−1.6 ± 0.8	−4.2 ± 1.0	−0.4 ± 0.8
TinyMWAMNet-S (MNv3-S, 128)	KD-std	−0.6 ± 0.7	−0.1 ± 0.1	−2.9 ± 1.6	−0.5 ± 0.6	−4.6 ± 2.1	0.7 ± 2.4
TinyMWAMNet-S (MNv3-S, 128)	CAD	−1.0 ± 0.4	−0.2 ± 0.2	−2.3 ± 1.7	−0.5 ± 0.5	−4.7 ± 3.0	−0.6 ± 1.7
MNv3-S, 160	CE-std	0.0 ± 0.5	0.0 ± 0.3	−2.4 ± 1.1	−2.2 ± 1.9	−3.8 ± 0.4	−1.1 ± 1.7
MNv3-S, 160	CAD	−0.3 ± 0.5	0.1 ± 0.4	−3.0 ± 2.2	−2.3 ± 0.7	−5.5 ± 3.3	0.4 ± 0.6
MNv2-0.35, 128	CE-std	−2.6 ± 0.9	−3.3 ± 0.9	−12.8 ± 1.8	−10.1 ± 1.0	−10.5 ± 1.9	−8.9 ± 1.4
MNv2-0.35, 128	CAD	−5.4 ± 2.8	−4.7 ± 1.8	−11.7 ± 1.1	−11.5 ± 0.6	−13.7 ± 2.5	−12.3 ± 2.1

Table S2: The size family: balanced accuracy (%, mean ± SD; three seeds for the size family, five for TinyMWAMNet-S and TinyMWAMNet-S6) in FP32 and after PTQ and QAT (with range warm-up). QAT failed for MobileNetV2-0.35 (Section 4.4 of the article).

		In-domain			**Shift-1**			**Shift-2**		
Model	**Regime**	**FP32**	**PTQ**	**QAT**	**FP32**	**PTQ**	**QAT**	**FP32**	**PTQ**	**QAT**
TinyMWAMNet-XS (MNv1-0.25, 96)	CE-std	97.8 ± 0.2	97.9 ± 0.3	98.1 ± 0.3	63.9 ± 1.0	63.9 ± 0.8	63.9 ± 0.6	74.1 ± 1.0	74.3 ± 0.9	73.8 ± 1.7
	CAD	98.0 ± 0.2	98.1 ± 0.2	98.0 ± 0.2	63.5 ± 0.5	63.4 ± 0.9	63.1 ± 0.5	76.0 ± 0.9	77.5 ± 1.0	75.9 ± 1.1
MNv3-S, 96	CE-std	98.6 ± 0.4	98.2 ± 0.5	98.3 ± 0.2	65.3 ± 0.7	62.8 ± 1.8	63.9 ± 0.7	76.5 ± 0.7	76.9 ± 1.9	76.3 ± 1.9
	CAD	98.4 ± 0.3	97.6 ± 0.7	98.5 ± 0.2	67.0 ± 0.6	63.9 ± 1.4	66.4 ± 0.5	82.7 ± 0.8	80.8 ± 1.4	82.5 ± 1.7
TinyMWAMNet-S (MNv3-S, 128)	CE-std	98.7 ± 0.2	98.3 ± 0.1	98.5 ± 0.5	67.2 ± 0.6	64.4 ± 1.8	65.7 ± 0.6	76.2 ± 0.9	76.0 ± 1.2	75.4 ± 0.8
	CAD	98.9 ± 0.0	98.2 ± 0.3	98.8 ± 0.2	67.2 ± 0.7	61.8 ± 1.4	66.6 ± 0.5	81.6 ± 0.9	77.0 ± 1.7	81.0 ± 1.9
MNv3-S, 160	CE-std	98.9 ± 0.3	98.0 ± 0.3	98.9 ± 0.4	66.4 ± 0.4	65.0 ± 1.0	64.1 ± 1.6	78.9 ± 3.4	75.9 ± 2.3	77.9 ± 2.2
	CAD	98.7 ± 0.3	98.0 ± 0.6	98.8 ± 0.3	67.9 ± 0.4	62.7 ± 0.8	65.6 ± 1.0	83.5 ± 1.2	77.7 ± 1.1	83.8 ± 1.3
MNv2-0.35, 128	CE-std	99.0 ± 0.3	99.0 ± 0.3	95.7 ± 0.7	66.6 ± 1.3	67.4 ± 1.4	56.4 ± 0.4	78.2 ± 1.6	77.4 ± 1.4	69.4 ± 0.3
	CAD	98.7 ± 0.3	98.6 ± 0.2	93.9 ± 1.9	66.2 ± 0.7	65.4 ± 2.1	54.6 ± 0.8	78.7 ± 0.7	77.7 ± 2.5	66.4 ± 2.6
TinyMWAMNet-S6 (ReLU6)	CE-std	94.7 ± 1.5	94.1 ± 1.6	94.2 ± 1.5	61.4 ± 1.4	60.9 ± 1.2	61.3 ± 1.5	73.3 ± 2.5	74.1 ± 1.9	72.7 ± 2.2
	CAD	96.7 ± 1.1	96.3 ± 1.2	96.6 ± 1.1	58.0 ± 0.7	58.0 ± 0.6	58.0 ± 0.7	76.4 ± 2.3	76.1 ± 3.3	75.6 ± 2.9

Table S3: Activation overflow after post-training quantization: INT8 activations at a clipping limit (per million activations) and images with at least one clipped activation (%), and, on the shifted sets, the overflow of images whose decision flipped and of images whose decision did not.

	Per million activations			**Images with any (%)**			**Shifted, per million**	
Model	**In-dom.**	**Shift-1**	**Shift-2**	**In-dom.**	**Shift-1**	**Shift-2**	**Flipped**	**Unchanged**
TinyMWAMNet-S (ReLU)	0.36	0.15	1.18	10.7	4.5	28.0	0.53	0.72
MNv3-S, 96 (ReLU)	0.76	0.24	1.38	11.4	4.3	20.3	0.51	0.87
MNv3-S, 160 (ReLU)	0.30	0.13	0.50	12.8	6.0	18.6	0.32	0.33
TinyMWAMNet-S6 (ReLU6)	2.41	0.77	0.76	30.4	18.2	19.9	0.53	0.78
TinyMWAMNet-XS (ReLU6)	34.34	20.39	20.04	74.6	62.3	64.0	3.84	21.00
MNv2-0.35 (ReLU6)	0.71	0.27	0.47	33.3	14.9	22.5	0.09	0.39
MWAMNet teacher, full INT8	3957.76	3975.43	4315.75	100.0	100.0	100.0	4295.02	4080.78

Table S4: Agreement between INT8 runtimes. Built-in: LiteRT's non-delegated kernels, used for every full-integer INT8 result in the paper except the "None" column of Table S1 (dynamic-range INT8 used XNNPACK). TFLM: the TensorFlow Lite Micro runtime, run through its Python interpreter with reference kernels. XNNPACK: LiteRT's default accelerated host kernels. Raw outputs are compared on 48 images per model (24 in-domain, 12 from each shifted set); XNNPACK decisions are compared on every evaluation image. Exact: share of identical `uint8` outputs; max $|\Delta|$: largest difference in output codes.

Comparison	**Models**	**Images**	**Decisions agree (%)**	**Worst model (%)**	**Exact (%)**	**Max $\lvert\Delta\rvert$**
Built-in vs TFLM (raw INT8 outputs)	12	576	99.13	95.8	77.4	92
Built-in vs XNNPACK (raw INT8 outputs)	12	576	98.78	95.8	72.6	92
Built-in vs XNNPACK (decisions, all images)	44	272,712	98.87	95.2	–	–

Table S5: Key models scored with TensorFlow Lite Micro (TFLM) on the full in-domain test set (380 images) and the 1,000-image subsets of each shifted set used for the teacher ladder (median seeds). S and S6: TinyMWAMNet-S and TinyMWAMNet-S6. BAcc: balanced accuracy (%); INT8 cost: INT8 minus FP32 balanced accuracy (points), scored with LiteRT's built-in kernels and with TFLM; Decisions and Exact: share of identical decisions and of identical `uint8` outputs between TFLM and the built-in kernels. No difference between TFLM and FP32 was significant after Holm correction over the 15 comparisons (McNemar; smallest uncorrected $p = 0.006$). Costs are computed from unrounded values.

		BAcc			**INT8 cost**		**TFLM vs built-in**	
Model	**Set**	**FP32**	**Built-in**	**TFLM**	**Built-in**	**TFLM**	**Decisions (%)**	**Exact (%)**
S, CAD, PTQ (seed 44)	In-domain	98.9	98.2	98.2	−0.8	−0.8	100.0	80.8
	Shift-1	66.0	62.4	63.0	−3.6	−3.0	99.1	48.6
	Shift-2	82.5	79.0	78.3	−3.5	−4.2	98.3	53.2
S, CAD, QAT (seed 44)	In-domain	98.9	98.9	98.9	0.0	0.0	100.0	87.6
	Shift-1	66.0	65.5	65.2	−0.6	−0.8	99.3	55.4
	Shift-2	82.5	81.0	80.3	−1.4	−2.2	98.8	59.5
S, CE-std, PTQ (seed 44)	In-domain	98.7	98.2	98.2	−0.5	−0.5	100.0	98.2
	Shift-1	64.9	61.1	61.2	−3.8	−3.7	99.2	88.9
	Shift-2	75.3	75.2	74.8	−0.1	−0.5	99.4	90.5
S, CE-std, QAT (seed 44)	In-domain	98.7	98.4	98.4	−0.3	−0.3	100.0	98.2
	Shift-1	64.9	63.4	63.3	−1.6	−1.6	98.7	87.4
	Shift-2	75.3	76.7	76.7	+1.4	+1.4	99.2	87.2
S6, CAD, PTQ (seed 46)	In-domain	96.8	96.8	96.8	0.0	0.0	100.0	83.4
	Shift-1	56.3	57.3	57.3	+1.0	+1.0	99.2	63.1
	Shift-2	73.9	72.3	72.5	−1.6	−1.4	99.4	62.1